Unit 7 Summary: Probability Distributions

Name _____

Random Variables and Distributions

Flip a coin 2 times. Let X = the number of heads.

x (heads)	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$

Idea	What it means	Example: flip a coin 2 times
Random variable	A number that comes from a chance process.	X = the number of heads.
X, x, and $P(x)$	Capital X is the random variable. Small x is one of its values. $P(x)$ means the probability that X equals x.	x is 0, 1, or 2. $P(1) = \frac{1}{2}$.
Probability distribution	It lists every value of X and its probability. It can be a table or a graph.	The table above.
Valid distribution	1. Every probability is from 0 to 1. 2. The probabilities add to 1: $\Sigma P(x) = 1$ (Σ means add up).	$\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. Valid.
Using a distribution	Add the probabilities of the values you want.	$P(X \geq 1) = P(1) + P(2)$ $= \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

Symbols: $\geq$ means "greater than or equal to." $\leq$ means "less than or equal to."

Theoretical or Empirical?

Kind	Where it comes from	Example
Theoretical	Reasoning about the chance process (equally likely outcomes, or the multiplication rule).	The coin flips above.
Empirical	Data. Divide each count by the total.	50 families: 0 cars: 5, 1 car: 20, 2 cars: 18, 3 cars: 7. So $P(0) = \frac{5}{50} = 0.10$, $P(1) = 0.40$, $P(2) = 0.36$, $P(3) = 0.14$.

Expected Value

The number of goals a soccer team scores in a game:

x (goals)	0	1	2	3	4
$P(x)$	0.1	0.3	0.3	0.2	0.1
$x \cdot P(x)$	0	0.3	0.6	0.6	0.4

Tool	**How it works**	**Example: goals in a game**
Expected value	$E(X) = \Sigma\, x \cdot P(x)$. $E(X)$ means the expected value of X. Multiply each value by its probability. Then add them all up.	$E(X) = 0 + 0.3 + 0.6 + 0.6 + 0.4$ = 1.9 goals
Mean of X	The expected value is also called the **mean** of X, written μ_X (μ is "mu").	μ_X = 1.9 goals
What it means	A **long-run average**. It tells you what happens over many tries. It does not tell you what happens on one try.	Over many games, the team scores about 1.9 goals per game on average. No single game has 1.9 goals.

Fair or Not?

	What it means	Example
Fair method of choosing	Every person has the same probability of being chosen.	Roll a number cube. 1 or 2: Ana, 3 or 4: Ben, 5 or 6: Cai. Each has $\frac{2}{6} = \frac{1}{3}$. **Fair.**
Not a fair method	Not every person has the same probability.	Flip 2 coins. HH: Ana, TT: Ben, one of each: Cai. Ana $\frac{1}{4}$, Ben $\frac{1}{4}$, Cai $\frac{2}{4} = \frac{1}{2}$. **Not fair.**
Fair game	The expected value of the money you win or lose is **0**. On average, no one wins or loses money.	Roll a number cube. You win \$3 if you roll a 6. You pay \$1 for any other number. $E(X) = 3 \cdot \frac{1}{6} + (-1) \cdot \frac{5}{6} = -\frac{2}{6} \approx -\$0.33$ You lose about 33 cents a game. **Not fair.**

Expected Payoff

A lottery ticket costs \$2. 1 in 1,000 tickets wins \$500. 1 in 50 tickets wins \$10. The rest win nothing.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win \$500	$500 - 2 = 498$	0.001	0.498
Win \$10	$10 - 2 = 8$	0.02	0.16
Win nothing	-2	0.979	-1.958
	$E(X) = \Sigma\, x \cdot P(x)$		-1.30

Tool	How it works	Example: the lottery ticket
Net gain	Net gain = what you win − what you paid. It is negative when you lose money.	Win \$500: $500 - 2 = 498$. Win nothing: -2.
Expected payoff	Let X = the net gain. Then $E(X) = \Sigma\, x \cdot P(x)$.	$0.498 + 0.16 + (-1.958) = -1.30$
What it means	The amount you win or lose on average.	On average, a buyer **loses \$1.30** per ticket. The lottery keeps that money.

Negative expected payoff: you lose money on average. **Expected payoff of 0:** the game is fair.

Protection Plans

Step	What to do	Example: a phone
Expected cost with no plan	Multiply the chance it breaks by the cost to replace it.	A phone costs \$300 to replace. The chance it breaks this year is 0.10. $0.10 \cdot 300 = \$30$
Compare	Compare it to the cost of the plan.	The plan costs \$40 for the year. On average, the plan costs \$10 more. That is how the company makes money.
Think about risk	Expected value is one part of a decision. Risk is another part: could you pay the big bill if it came?	The plan protects you from a sudden \$300 bill. If you could not pay it, the plan might be worth it to you.